Fiscal Year:	FY 2010	**Task Last Updated:**	FY 11/12/2010
PI Name:	Cisar, Alan Ph.D.		
Project Title:	Portable Cathode-Air-Vapor-Feed Electrochemical Medical Oxygen Concentrator		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	(1) **Medical Conditions**:Risk of Adverse Health Outcomes and Decrements in Performance Due to Medical Conditions that occur in Mission, as well as Long Term Health Outcomes Due to Mission Exposures		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	alan.cisar@lynntech.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	(979) 693-0017
Organization Name:	Lynntech, Inc.		
PI Address 1:	7610 Eastmark Drive		
PI Address 2:			
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77840-4023	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	02/05/2010	**End Date:**	02/08/2013
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Watkins, Sharmi1a	**Contact Phone:**	281.483.0395
Contact Email:	sharmila.watkins@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date change to 02/08/2013 per 3/14/2012 HRP Master Task List information (Ed., 3/28/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX10CA96C		
Performance Goal No.:			
Performance Goal Text:			
	Future space exploration missions present significant new challenges to crew health care capabilities, particularly in the efficient utilization of on-board oxygen resources. The International Space Station and future exploration vehicles require a light weight, compact, portable oxygen concentrator technology (OCT) that can provide medical grade oxygen from the ambient cabin air. Current OCTs are heavy, bulky, have a narrow operating temperature range (ambient to 40 degree Celsius), and require 15 to 30 minutes start-up time to reach their full operating capacity. Lynntech's proposed electrochemical OCT solves these issues by operating the OCT with a cathode-air vapor feed, unlike conventional electrochemical OCTs which require a liquid water feed. This is possible due to the use of in-house developed proprietary nanocomposite proton exchange membrane and catalyst technologies. Cathode-air vapor feed operation eliminates the need for a bulky on-board water supply, significantly reduces the complexity of the balance-of-plant, and		

Task Description:	greatly increases the system efficiency. OCT will be a quarter the size and weight of conventional OCTs, be capable of instant start-up, and have a wide operating temperature range. Lynntech will develop a Phase II prototype that is capable of delivering 4 SLPM of 60% oxygen and deliver it to NASA for further testing. POTENTIAL NASA COMMERCIAL APPLICATIONS : Lynntech's lightweight, portable electrochemical oxygen concentrator technology has the potential to provide a compact, robust, on-demand/continuous oxygen concentrating device. Providing lightweight compact oxygen concentrator devices will have applications for NASA, ISS and future exploration vehicles supporting long duration missions that require improved medical capabilities and enhanced on-board resource utilization, particularly oxygen delivery. This technology will enable both ambient pressure operating devices (for portable applications) and pressurized (up to 3600 psia) oxygen concentrator devices (for stationary applications). With minor design changes, this OC system can also be used as high pressure oxygen compressor (up to 3600 psi).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A lightweight, portable and/or mobile, oxygen concentrator, and one, in particular, that is capable of delivering minimum 60% humidified oxygen gas at pressures of 5 psi and above, would find many applications in the health services industry. It is estimated that 2 to 4 million cardiopulmonary patients in the U.S. require supplementary oxygen; approximately 500,000 of which require continuous oxygen therapy. Current means to supply oxygen include compressed oxygen cylinders (heavy and expensive); liquid oxygen (cryogenic and expensive); and molecular sieve-bed oxygen concentrators that use molecular sieve technology. Weight, size, power, and cost requirements for current commercially available oxygen concentrators have been the major hurdles for gaining the market acceptance, thus limited availability to those in need of oxygen therapy. With Lynntech's advanced electrochemical oxygen concentrator technology, this need can be satisfied very easily to improve the quality of lifestyle of those in need of a portable, compact oxygen generator.
Task Progress:	New project for FY2010. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated:)